Roll No. CS-112

NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD

FACULTY OF ENGINEERING & COMPUTING

DEPARTMENT OF COMPUTER SCIENCE

END-TERM EXAMINATION – SPRING 2025 – BSCS VII (AFTERNOON)

Subject: Computer Graphics | **Instructor: Sadia Ashraf**

Allowed Time: 03 Hours | **Total Marks: 50**

Instructions: -

I. Attempt all questions and return the question paper with the answer sheet.

II. Clearly mention the question number before attempting it

QUESTION 1 [CLO-2, C3, PLO-3]

Solve the answer for the following briefly. **[5+5+5+5 Marks]**

a) Describe the key difference between perspective and parallel projection. Draw simple diagrams to illustrate.

b) A window is defined in world coordinates as (xmin=10, ymin=20, xmax=100, ymax=200), and the viewport is defined as (vxmin=0, vymin=0, vxmax=500, vymax=600). Write the formula and compute the screen coordinates for a world point (50, 60).

c) For a region surrounded by black pixels, write code to use flood fill to color the interior using green. Assume the starting point is inside the region.

d) A user draws a line from (0, 50) to (150, 50). If the clipping window is (10, 10) to (100, 100), what will be the final displayed segment?

QUESTION 2 [CLO-3, C3, PLO-4]

Construct pseudocode to implement Cohen-Sutherland line clipping and apply it to clip the line from (0, 0) to (120, 120) for a window of (10, 10) to (100, 100). **[10 Marks]**

QUESTION 3 [CLO-3, C3, PLO-4]

Apply Bresenham's to find intermediate points for a line that has the following Starting and Ending points A(2, 3) B(13, 16). **[10 Marks]**

QUESTION 4 [CLO-3, C3, PLO-4]

Apply parallel projection to the given image and create 6 multi View projections for the 3d model provided. **[10 Marks]**